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Pretreatment with alcoholic extract of *Crataegus oxycantha* (AEC) activates mitochondrial protection during isoproterenol – induced myocardial infarction in rats

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Abstract

Crataegus oxycantha (hawthorn) is used in herbal and homeopathic medicine as a cardiotonic. The present study was done to investigate the effect of the alcoholic extract of *Crataegus oxycantha* (AEC) on mitochondrial function during experimentally induced myocardial infarction in rat. AEC was administered orally to male albino rats (150–200 g), at a dosage of 0.5 ml/100 g body weight/day, for 30 days. At the end of the experimental period, the animals were administered isoproterenol (85 mg/kg body weight, s.c) for 2 days at an interval of 24 h. After 48 h, the rats were anaesthetized and sacrificed. The hearts were homogenized for biochemical and electron microscopic analysis. AEC pretreatment maintained mitochondrial antioxidant status, prevented mitochondrial lipid peroxidative damage and decrease in Kreb's cycle enzymes induced by isoproterenol in rat heart. (Mol Cell Biochem **292:** 59–67, 2006)

Key words: alcoholic extract of *Crataegus*, mitochondria, lipid peroxidation, antioxidant, isoproterenol

Introduction

Myocardial infarction is the primary cause of mortality in developed and developing nations. Myocardial infarction usually results from abrupt reduction in coronary blood flow to a segment of the myocardium, which initiates a continuum of progressively more severe cellular changes that, unless interrupted by early reperfusion, inevitably culminate in cell death and tissue necrosis [1].

Evidence indicates that mitochondria contribute to reperfusion-induced increase in the production of free radicals [2]. Superoxide dismutase (SOD) is an effective protein for self-preservation because it scavenges free radicals in reperfusion injury. Therefore, many studies have been done to investigate whether SOD could be a myocardial protectant [3–6]. Mammals have three distinct SODs: copper and zinc SOD (CuZnSOD), manganese SOD (MnSOD) and extracellular SOD [7, 8]. Of them, MnSOD reportedly exists abundantly in mitochondria to attenuate reperfusion injury by dismutating the superoxide occurring in mitochondria [3, 9].

Herbal drugs are receiving much attention in the treatment of many diseases, based on their non-toxic nature and natural occurrence. Recent scientific interest is also focused on the identification of naturally occurring compounds with cardioprotective effect. C*rataegus oxycantha* is one such medicinal plant which is used as a cardiotonic. Alcoholic extract of the berries of *Crataegus oxycantha* (AEC) (Hawthorn), L. Rosaceae, is used in herbal medicine as a heart tonic [10]. AEC has a number of pharmacological properties and this is possibly due to its major constituents-flavonoids, triterpene saponins and some cardioactive amines [11]. It contains a large amount of flavones, which possess

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significant antioxidant activity. The primary cardioprotective activity of this plant is generally attributed to its oligomeric proanthocyanidins (OPCs) [12]. The hypolipidemic and antiatherosclerotic effect of AEC have been reported from this lab, in rats given a high cholesterol diet [13]. AEC prevents the defective oxidative phosphorylation and diminished energy production in isoproterenol-induced myocardial infarction [14]. We have also shown that AEC protected the liver damage caused by isoproterenol induction [15]. However, the mitochondrial protection rendered by AEC against myocardial infarction has not been investigated. The present study was designed to investigate the effect of AEC on mitochondrial Kreb's cycle enzymes, lipid peroxidative damage, antioxidant status and ultrastructural changes during isoproterenol-induced myocardial infarction in rat heart.

Materials and methods

Experimental procedure

Male albino rats of Wistar strain, weighing approximately 150–200 g, were obtained from the Tamilnadu University of Veterinary and Animal Sciences, Madhavaram, Chennai, India. The animals (Ethical clearance obtained from the Institutional Animal Ethics Committee – IAEC No: – 01/008/02) were housed in solid-bottomed polypropylene cages and received commercial rat chow (Hindustan Lever Ltd, Bangalore, India) and water, *ad libitum*. The animals were divided into five groups with six rats in each group.

The berries of *Crataegus oxycantha* were collected from Western Himalayas, and obtained through Bahola Industries Pvt Ltd, Chennai, India. The extract was prepared by mixing 100 g of fresh pulp of the fruits. The berries of *Crataegus oxycantha* was defatted by soaking in petroleum ether (60–80 $^\circ$) for 14 days and the solvent removed at low pressure. To the defatted powder, 95% ethanol was added and the extraction was done by cool percolation method. The extract was concentrated in a water bath under reduced pressure.

Group I – Control animals received normal rat diet and water, *ad libitum*.
Group II – Drug control, received normal rat diet and were given AEC (0.5 ml/100 g.b.wt.) [10], orally for 30 days.
Group III – Isoproterenol (85 mg/kg.b.wt. Sigma Chemical Co, USA) was given by subcutaneous injection for 2 days at an interval of 24 hours (31st and 32nd day) [16].
Group IV – AEC was given orally for 30 days. Isoproterenol was given by s.c injection on 31st and 32nd day.
Group V – Captopril was given orally for 30 days. Isoproterenol was given by s.c injection on 31st and 32nd day. Captopril was used as a positive control [17].

After the experimental period, the animals were sacrificed by ether anesthesia and decapitation. Heart was dissected out immediately, one portion of heart was used for electron microscopic examination and the other portion was used for the isolation of mitochondria.

Isolation of mitochondria

The heart was excised, rinsed in ice-cold isotonic saline, blotted with filter paper, weighed, homogenized in 0.25 M sucrose at 4 $^\circ$C and mitochondria were isolated by the method of Johnson and Lardy [18]. Briefly, Immediately after sacrifice, the liver was removed and all the blood vessels and connective tissues were trimmed off. The tissue was washed twice with the ice-cold buffer (Sucrose: 0.25 M, Tris-HCl: 5 mM, pH 7.4, EDTA: 1 mM) and transferred to a 50 ml beaker, minced into small sections with scissors. 10 ml of cold buffer was used as the homogenizing medium and homogenate was prepared by approximately six strokes.

The homogenate was centrifuged in a refrigerated high-speed centrifuge at 2000 rpm for 10 min. The pellet containing the nucleus, red blood cells and cell debris was discarded and the supernatant was spun at 10,000 rpm for 7 min. The pellet contains the mitochondrial fraction.

In appearance the pellet had 3 distinct regions.

The middle layer, which was medium to dark brown in colour and contained intact mitochondria was carefully removed and resuspended in 10 ml of the cold buffer and again centrifuged at 10,000 rpm for 7 min. The resulting mitochondrial pellet was again washed in 1.15% potassium chloride, centrifuged at 10,000 rpm for 10 min. The final pellet was suspended in 1.15% potassium chloride, gently homogenised and used as the enzyme source.

Biochemical parameters

Kreb's cycle enzymes
The activity of Isocitrate dehydrogenase was estimated by the method of King [19] using trisodium isocitrate as substrate and the activity was expressed as nmoles of α- ketoglutarate produced/h/mg protein. α-ketoglutarate dehydrogenase was assayed by the method of Reed and Mukherjee [20] using potassium α-ketoglutarate as the substrate and the activity was expressed as nmoles potassium ferrocyanide liberated/h/mg protein. Similarly, malate dehydrogenase [21], succinate dehydrogenase [22], NADH dehydrogenase [23] and cytochrome-c-oxidase [24] were assayed using oxaloacetate, succinate, NADH and cytochrome-c-oxidase as substrates, respectively, and the activities were expressed as μmoles of NADH oxidized/min/mg protein, μmoles of succinate oxidized /min/mg protein, μmoles of

Table 1. Activities of mitochondrial Kreb's cycle enzymes in the heart of control and experimental animals

Parameters	Group I	Group II	Group III	Group IV	Group V
Isocitrate dehydrogenase[a]	764.6 ± 1.5	757.4 ± 3.2*	537.0 ± 1.9*	657.7 ± 1.2*#$	672.4 ± 1.0*#
α-Keto glutarate dehydrogenase[b]	80.1 ± 0.6	76.9 ± 1.4*	61.3 ± 0.8*	66.6 ± 1.2*#$	67.3 ± 0.8*#
Malate dehydrogenase[c]	327.9 ± 0.9	328.2 ± 1.9*	159.7 ± 1.1*	251.4 ± 1.2*#$	205.9 ± 1.1*#
Succinate dehydrogenase[d]	43.4 ± 1.0	43.6 ± 1.2*	15.7 ± 0.3*	35.7 ± 0.7*#$	34.7 ± 1.1*#
NADH dehydrogenase[c]	32.5 ± 0.9	31.3 ± 1.2*	13.1 ± 0.3*	28.9 ± 1.3*#$	28.3 ± 0.5*#
Cytochrome C oxidase[e]	5.2 ± 0.3	4.8 ± 0.2*	2.9 ± 0.2*	4.2 ± 0.1*#$	4.3 ± 0.3*#

[a]nmoles of αkg/h/mg protein.
[b]nmoles of ferricyanide/h/mg protein.
[c]μmoles of NADH oxidized/min/mg protein.
[d]μmoles of succinate oxidized/min/mg protein.
[e] O.D × 10^{-2}/min/mg protein.
Values are expressed as mean ± SEM.
$P < 0.05$, * Compared with Group I, # Compared with Group I and Group III, $ Compared with Group V.

NADH oxidized/min/mg protein and O.D × 10^{-2}/min/mg protein, respectively.

Mitochondrial lipid peroxidation
Levels of mitochondrial lipid peroxidation products were estimated by the method of Santos *et al.* [25] and expressed as nanomoles of malondialdehyde (MDA) formed/mg protein.

Mitochondrial antioxidants
The activity of glutathione reductase was assayed by the method of Staal *et al.*, [26] using oxidized glutathione as substrate and it was expressed as nmoles NADPH oxidized/min/mg protein, Similarly, glutathione-S-transferase [27], glutathione peroxidase [28], superoxide dismutase [29] and catalase [30] were assayed using reduced glutathione, pyrogallol and H_2O_2, respectively, as substrates and the activities were expressed as nmoles Chloro dinitrobenzene (CDNB) conjugated/min/mg protein, nmoles of glutathione consumed/min/mg protein, units/mg protein and nmoles of H_2O_2 released/min/mg protein, respectively.

Mitochondrial MnSOD
MnSOD activity was determined using activity staining in native Polyacrylamide gel as described by Beauchamp and Fridovich [31]. Extracts containing 40 μg mitochondrial proteins were separated on a 12% native polyacrylamide gel [32]. The separated SOD was stained with Riboflavin – 1.52 mg, Nitroblue tetrazolium – 20.4 mg, EDTA – 37.2 mg, TEMED – 324 μl in 100 ml of 50 mM potassium phosphate buffer (pH 7.8). The activity of MnSOD was quantified by densitometry. (Camag Scanner, Switzerland).

Ultrastructural examination
The hearts were removed, washed immediately with 3% glutaraldehyde solution and then fixed with the same. These sections were then examined under an electron microscope (CM-12, Philips technologies, operating at 80 KV).

Statistical analysis
Values are expressed as mean ± SEM. The results were statistically evaluated using one-way analysis of variance (ANOVA) by SPSS software for repeated measurements and, when indicated, a post hoc Tukey's test for multiple comparisons. The mean difference is significant at the 0.05 level.

Results

Kreb's cycle enzymes (Table 1)

Isocitrate dehydrogenase, α-ketoglutarate dehydrogenase, malate dehydrogenase, succinate dehydrogenase, NADH dehydrogenase and cytochrome-c-oxidase were significantly decreased in isoproterenol treated (Group III) rats when compared to control (Group I) and drug control (Group II) rats. The activities of Kreb's cycle enzymes were significantly increased in AEC pretreated (Group IV) and captopril pretreated (Group V) rats when compared with isoproterenol administered (group III) animals.

Mitochondrial lipid peroxidation (Fig. 1)

The levels of mitochondrial lipid peroxides were significantly increased in isoproterenol-induced (Group III) rats when compared to control (Group I) rats. AEC and captopril

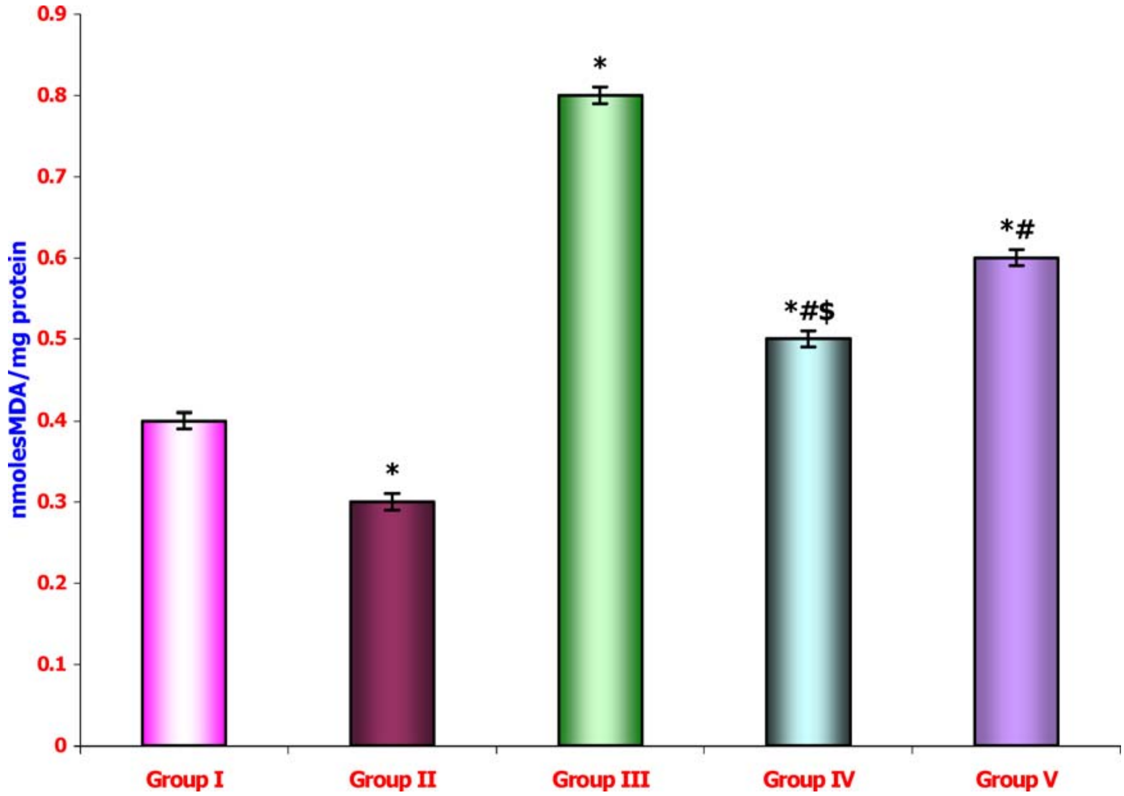

Fig. 1. Levels of heart mitochondrial lipid peroxides in control and experimental animals, Values are expressed as mean ± SEM, $P < 0.05$, *Compared with Group I, # Compared with Group I and III, $ Compared with Group V.

pretreated (Group IV and V) rats showed significantly decreased levels of heart mitochondrial lipid peroxides when compared with group III rats.

Mitochondrial antioxidants (Table 2)

All the antioxidant enzymes (Superoxide dismutase, catalase, glutathione-S-transferase, glutathione peroxidase, glutathione reductase) and reduced glutathione were significantly decreased in isoproterenol-induced (Group III) animals when compared with control and drug control (Group I and Group II, respectively) animals. AEC (Group IV) and captopril treated (Group V) animals showed a significant increase in the activities of antioxidant enzymes when compared with myocardially infarcted Group III animals.

Mitochondrial MnSOD (Plate 1)

Control and AEC alone treated rats showed normal levels of MnSOD activity. Densitometric scanning results showed that in Isoproterenol-induced animals, the MnSOD activity (1390.6 ± 78.8) was significantly decreased when compared to controls (2437.9 ± 117.2) and AEC alone (6537.7 ± 251.5) treated rats. Mitochondrial MnSOD was expressed at higher levels in AEC pretreated group (9165.5 ± 130.1) when compared with Isoproterenol-administered group.

Ultrastructural studies (Plate 2)

Control rat heart showed normal architecture without any disruption of mitochondria (15000×) (Plate 2a). Isoproterenol - administered rat heart showed mitochondrial swelling and disruption of mitochondrial cristae (15000×) (Plate 2b). In AEC pre treated rat heart (Plate 2c), mitochondrial swelling was reduced (15000×). AEC treatment (Plate 2c) protected the mitochondria against isoproterenol-induced (Plate 2b) myocardial infarction in rat.

Discussion

Mitochondria play a central role in the energy-generating process within the cell. Apart from this important function, mitochondria are involved in complex processes such as apoptosis. The cardioprotective actions of potassium channel

Table 2. Activities of mitochondrial antioxidant enzymes in the heart of control and experimental animals

Parameters	Group I	Group II	Group III	Group IV	Group V
Superoxide dismutase[a]	20.7 ± 0.7	$21.5 \pm 0.5^{*}$	$14.7 \pm 0.4^{*}$	$17.5 \pm 0.3^{*\#\$}$	$16.9 \pm 0.3^{*\#}$
Catalase[b]	2.5 ± 0.1	$2.4 \pm 0.06^{*}$	$0.6 \pm 0.02^{*}$	$1.6 \pm 0.04^{*\#\$}$	$1.5 \pm 0.03^{*\#}$
Glutathione – S – transferase[c]	69.6 ± 0.7	$68.0 \pm 0.8^{*}$	$39.6 \pm 0.6^{*}$	$55.0 \pm 1.2^{*\#\$}$	$52.2 \pm 0.8^{*\#}$
Glutathione peroxidase[d]	1.3 ± 0.06	$1.2 \pm 0.05^{*}$	$0.9 \pm 0.06^{*}$	$1.1 \pm 0.06^{*\#\$}$	$1.2 \pm 0.05^{*\#}$
Glutathione reductase[e]	8.4 ± 0.08	$8.9 \pm 0.05^{*}$	$6.2 \pm 0.08^{*}$	$8.8 \pm 0.2^{*\#\$}$	$7.9 \pm 0.06^{*\#}$
Reduced Glutathione[f]	7.17 ± 0.65	$7.19 \pm 0.68^{*}$	$4.36 \pm 0.41^{*}$	$5.61 \pm 0.58^{*\#\$}$	$5.50 \pm 0.48^{*\#}$

[a]units/mg protein.
[b]nmoles of H_2O_2 released/min/mg protein.
[c]nmoles of CDNB conjugated/min/mg protein.
[d]nmoles of GSH oxidized/ min/mg protein.
[e]nmoles of NADH oxidized/min/mg protein.
[f]nmoles of GSH reduced/mg protein.
Values are expressed as mean ± SEM.
$P < 0.05$, * Compared with Group I, # Compared with Group I and Group III, \$ Compared with Group V.

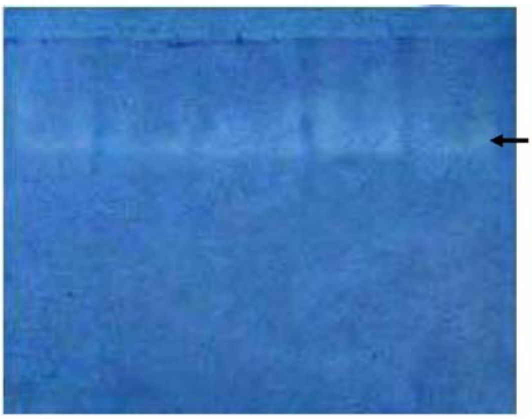

Plate 1. Heart mitochondrial MnSOD expression in control and experimental animals (Zymogram of MnSOD).

openers have revealed that cardiac mitochondria are more important as the primary targets of these drugs than the plasma membrane [33].

Decreases in the activities of Kreb's cycle enzymes in isoproterenol administered rats could be due to enhanced phospholipid degradation resulting in the non-availability of cardiolipin for their functional activity. Cytochrome c directly activates downstream effectors when injected into the cytosol in the absence of upstream signals. The dehydrogenases of TCA cycle could have been affected by the free radicals produced on isoproterenol treatment. AEC pretreatment protects the heart against mitochondrial oxidative stress induced by isoproterenol by enhancing the activities of Kreb's cycle enzymes.

The electron micrograph further confirms the damage to mitochondria, on isoproterenol treatment, where lamellae of mitochondria were lost. Enzymes involved in energy production are associated with the mitochondrial membrane and the isoproterenol-induced damage to mitochondria reduced the activities of Kreb's cycle enzymes.

Mitochondrial membrane contains relatively large amounts of polyunsaturated fatty acids in its phospholipids which are highly susceptible to lipid peroxidation, an important deterioration in biological membrane [34]. Isoproterenol induces lipid peroxidation of mitochondrial membrane [35]. The present study also shows that mitochondrial lipid peroxidative products were increased during isoproterenol administration (Fig. 1). AEC pretreatment protected the heart mitochondrial membrane against lipid peroxidative damage by reducing the isoproterenol-induced lipid peroxidation. AEC administration prevented the decrease in the activities of mitochondrial membrane associated enzymes in pretreated animals and this effect is probably due to removal of excess free radicals generated by isoproterenol.

The main constituents of AEC are amines (β-phenethylamine, tyramine, acetyl choline) triterpene saponins (Oleanolic acid, ursolic acid, crataegolic acid), flavonoids and their glycosides (quercetin, hyperoside, rutin, vitexin), monomeric catechins and Oligomeric proanthocyandines (OPCs) [36].

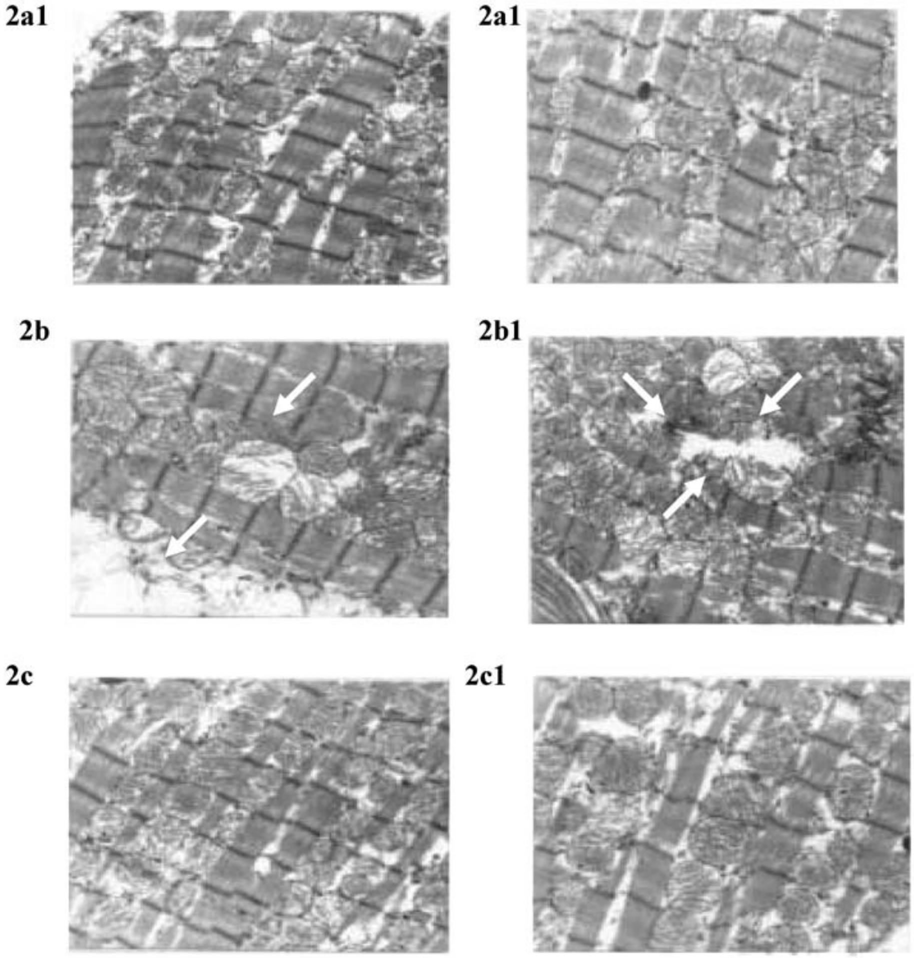

Plate 2a & 2a1 - Control rat heart showing normal architecture without any disruption of mitochondria (15000x)

Plate 2b & 2b1 - Isoproterenol - administered rat heart showing mitochondrial swelling and disruption of mitochondrial cristae (15000x)

Plate 2c & 2c1 - AEC pretreated rat heart showing reduced mitochondrial swelling (15000x)

Plate 2. Ultrastructural studies in heart mitochondria of control and experimental animals.

Flavonoids have been shown to inhibit lipid peroxide formation in rat tissue and also inhibit the free radical production in cells at various stages. They can interact with superoxide anions chelate metal ions and react with peroxy radicals [37]. The flavonoids in AEC probably protected the heart from myocardial damage by scavenging free radicals and thereby suppressing the peroxidation of lipids. The triterpene saponins, ursolic and oleanolic acid, present in AEC may also contribute to the antioxidant effect of AEC, since they also exhibit remarkable antioxidant activities [38].

GSH, a tripeptide, plays a crucial role in cell viability. GSH, either alone, or, in conjugation with other proteins, can protect cells against lipid peroxidation [39]. Mitochondrial GSH plays a critical role in maintaining cell viability through the regulation of mitochondrial inner membrane permeability by maintaining sulphydryl groups in the reduced state. Mitochondria do not contain the enzymes for GSH biosynthesis [40] and hence the mitochondrial pool of GSH is derived from the cytosolic GSH which is transported into mitochondrial matrix [41, 42].

Glutathione (GSH), together with glutathione-dependent enzymes, glutathione peroxidase, glutathione-S-transferase (GST), glutathione reductase, catalase and superoxide dismutase (SOD), efficiently scavenge toxic free radicals [43]. GPx, an antioxidant enzyme, offers protection to the mitochondrial membrane from peroxidative damage. Under conditions of severe oxidative stress, GPx is inactivated [44]. Adequate supplies of glutathione and NADPH are required for the activity of GPx and reduced availability of glutathione as observed in isoproterenol-administered rats (Table 2) and NADPH resulted in decreased activity of GPx [45]. A decrease in the activity of GPx (Table 2) makes mitochondria susceptible to isoproterenol induced myocardial damage, which leads to a change in mitochondrial function. Similarly, the reduced availability of glutathione leads to decrease in the activity of GST (Table 2) in isoproterenol-administered rats. AEC pretreatment maintained the activity of GPx and GST (Table 2).

Glutathione reductase and GPx are essential for maintaining the constant ratio of reduced glutathione to oxidized glutathione in the cell. Inactivation of glutathione reductase in heart as reported in Table 2 leads to accumulation of GSSG, the oxidized product of GSH [2], which inactivates enzymes containing - SH groups and inhibits protein synthesis [46].

Mitochondria are the main source of the superoxide radical and other reactive oxygen species that may be generated from them [47]. The main mechanisms responsible for mitochondrial ROS production are the respiratory chain, in particular its complexes I and III [48–50] in the inner mitochondrial membrane and monoamine oxidase in the outer membrane. To protect mitochondria and the cell against the damaging effects of ROS, several measures can be applied. One of them is increasing the intracellular glutathione content. This can be done by supplying precursors for glutathione synthesis. In the present study, AEC pretreatment was found to decreasing the lipid peroxidative damages induced by isoproterenol.

SOD is a class of enzymes, which catalyzes the dismutation of two superoxide radicals to form hydrogen peroxide and molecular oxygen [51]. Manganese containing mitochondrial superoxide dismutase, located in the matrix, eliminates $\bullet O_2^-$ generated by misfires in the electron transport chain, by catalyzing dismutation to H_2O_2. H_2O_2 is inactivated by either catalase [52] or by the GSH redox system consisting of reduced GSH as the cofactor for GPx and GSH reductase [53]. The relative contributions of catalase [52] and GPx in H_2O_2 degradation in cardiac mitochondria remain unclear. These enzymes serve to minimize the accumulation of $\bullet O_2^-$ and H_2O_2 [54]. Modulation of the mitochondrial SOD isoforms is a critical determinant in the tolerance of the heart to oxidative stress. MnSOD activity and content decreases after ischemia, presumably via inactivation/degradation of mature, active MnSOD within mitochondria [55]. Similar trends were observed in the present study; the mitochondrial MnSOD content was decreased on isoproterenol administration.

Overexpression of mitochondrial SOD at a targeted intracellular location protects the heart against ischemic injury [56]. The present results suggest that overexpression of MnSOD (Plate 1) on AEC pretreatment in isoproterenol administered rats scavenges the superoxide radicals and thus prevents the formation of other more toxic radicals such as hydroxy radicals and peroxy radicals. It is well known that superoxide anions can participate in the metal ion-mediated Haber-Weiss reaction that converts hydrogen peroxide to the very toxic hydroxy radicals [56]. AEC treatment protected the mitochondria against isoproterenol- induced depletion of MnSOD. The results indicate that AEC is comparable to captopril as a cardioprotectant.

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